MILITARY SPECIFICATION

PRIMER AND BASE, IGNITER, XM110, PARTS FOR AND LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the parts, loading, assembling and packing for Primer and Base, Igniter, XM110, for use with Igniter, Time, Blasting Fuze, Weatherproof, M60.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	- Preservation, Methods of
MIL-A-2550	- Ammunition and Special Weapons; General Specification for.
MIL-P-12951	- Primer, Percussion, M39A1, Loading, Assembling and Packing
MIL-I-45607	- Inspection Equipment, Supply and Maintenance of.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105).
MIL-STD-109	- Quality Assurance Terms and Definitions.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

FSC: 1375

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:39Z
Check the source to verify that this is the current version before use.

DRAWINGS

U. S. ARMY MUNITIONS COMMAND

8822515 - Primer and Base, Igniter, XM110
7548103 - Box, Set-up, Packing, Ammunition, Outer for Small Explosive Components.
7548104 - Box, Set-up, Packing, Ammunition, Inner for Small Explosive Components.
9224760 - Box, Packing, Ammunition for Primer Base Assembly, XM110.

PUBLICATIONS

U. S. ARMY MUNITIONS COMMAND

ET 8822515 - Equipment Tabulation Number

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

3. REQUIREMENTS

3.1 Material - Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Primer and Base Assembly - The primer and base assembly shall comply with all requirements specified on Drawing (dwg.) 8822515 and with all requirements specified on applicable specifications.

3.3 Functioning

3.3.1 Duds - The primer and base assembly shall not fail to function on first attempt and no primer shall show evidence of a hangfire or squibbing (see 6.5) when tested as specified in 4.4.1.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:39Z
Check the source to verify that this is the current version before use.

3.4 Workmanship - All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, chips, scale, dirt, grease, rust, salt deposits and other foreign material. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

3.5 First Article Inspection - This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define the terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product - At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:39Z
Check the source to verify that this is the current version before use.

a. A statement that the lot complies with all quality assurance provisions specified within this specification.
b. Number of units of product inspected.
c. Results obtained, by defect code, for all inspections performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.
f. Number of items in the lot.
g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 First Article Inspection

4.2.1 Submission - The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of one hundred (100) assemblies and fifty (50) complete sets of parts (i.e. 50 each of every component and every subassembly) which have been produced by the contractor or

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:39Z
Check the source to verify that this is the current version before use.

furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the sime source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials, (see 4.1.1e). A sample containing known defects will not be submitted unless specifically authorized, by the Contracting Officer. A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be performed - All assemblies and components will be inspected by the Government for all dimensions and requirements referenced on the applicable drawings, and for all requirements, examinations, and tests specified in the applicable specifications. Testing shall be as follows:

4.2.2.1 Functioning (see 3.3) - Sample size - 100 assemblies. The primer and base assembly shall be tested as specified in 4.4.1.

4.2.3 Rejection - If any assembly or component fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component to comply with any of the stated requirements.

4.3 Inspection Provisions

4.3.1 Lot formation - The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:39Z
Check the source to verify that this is the current version before use.

represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, specification, or revision thereof. Changes to either the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government inspector, shall not be deemed to alter the homogeneity of the lot. Each lot shall contain:

a. Primers bearing the same lot interfix number from one manufacturer.

b. Parts of one kind from one supplier.

Inspection lots shall comply with MIL-STD-105 and shall be numbered in accordance with MIL-STD-1168.

Unless otherwise approved by the contracting officer, the inspection lot size of major assemblies or end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest quantity contractually scheduled for delivery during any month of the contract period. Inspection lots for components or sub-assemblies, other than the items of delivery, shall be homogeneous and of a size convenient to the contractor.

4.3.2 Examination - Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects (and Majors, when so specified) shall be 100 percent. Contractor's sampling plans, if used, shall be approved by the Government and shall provide, as a minimum, the protection afforded the Government by the sampling plans in Military Standard MIL-STD-105. Continuous sampling plans in accordance with Standard MIL-STD-1235 for defects other than critical may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:39Z
Check the source to verify that this is the current version before use.

4.3.2.1 Primer (see dwg. 8798919 covering a detail of dwg. 8822515).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Primer damaged	Visual	01001
Minor: None defined			

4.3.2.2 Primer base (see dwg. 8822516 covering a detail of dwg. 8822515).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.65 percent		
101.	Diameter of primer base hole	Gage	02001
102.	Base flash hole obstructed	Visual	02002
103.	Base split or cracked	Visual	02003
Minor:	AQL 1.50 percent		
201.	Total length	Gage	02004
202.	Large outside diameter	Gage	02005
203.	Small outside diameter	Gage	02006
204.	Evidence of poor workmanship (see 3.4)	Visual	02007

4.3.2.3 Primer and Base Assembly (see dwg. 8822515).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Primer not fully seated	Visual	03001

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:39Z
Check the source to verify that this is the current version before use.

Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.4).....	Visual	03002

4.3.2.4 Box, Set-up, Packing, Ammunition, Inner (see dwg. 7548104).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Incorrect number of assemblies in box.....	Visual	04001
102.	Box damaged exposing contents....	Visual	04002
Minor:	AQL 1.50 percent		
201.	Marking misleading or unidentifiable.....	Visual	04003
202.	Any packing material missing.....	Visual	04004
203.	Spacer separator damaged to extent that assemblies touch each other.....	Visual	04005

4.3.2.5 Box, Set-up, Packing, Ammunition, Outer (see dwg. 7548103).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Number of inner set-up boxes incorrect.....	Visual	05001
102.	Sealing strip torn, wrinkled, incomplete, or otherwise fails to seal box completely.....	Visual	05002

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:39Z
Check the source to verify that this is the current version before use.

Minor:	AQL 1.50 percent		
201.	Box cut or contains hole.....	Visual	05003
202.	Contents loose.....	Manual	05004
203.	Marking misleading or unidentifiable.....	Visual	05005

4.3.2.6 Box, Set-up, Packing, Ammunition, Outer in sealed barrier bag (see dwg. 7548103).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Seal missing, incomplete or incorrectly located.....	Visual	06001
102.	Bag punctured, torn or cut.....	Visual	06002
Minor:	AQL 0.65 percent		
201.	Marking misleading or unidentifiable.....	Visual	06003

4.3.2.7 Box, Packing, Ammunition (see dwg. 9224760).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Box damaged to extent that contents are exposed.....	Visual	07001
102.	Strapping missing or loose.....	Visual	07002
Minor:	AQL 1.50 percent		
201.	Marking misleading or unidentifiable.....	Visual	07003
202.	Strapping improperly assembled...	Visual	07004
203.	Contents loose.....	Manual	07005

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:39Z
Check the source to verify that this is the current version before use.

4.3.3 Testing

4.3.3.1 Functioning

4.3.3.1.1 Duds (see 3.3.1) - Major defect - Code No. 08001.- A sample of 80 primer base assemblies shall be selected from each lot for these tests. The tests shall be performed as specified in 4.4.1.1 using test equipment specified in 4.3.4. If two or more defectives are found the lot shall be rejected. If one defective is found a second sample of 80 primer base assemblies shall be tested. If the total number of defectives in both the first and second sample totals two or more, the lot shall be rejected.

4.3.3.2 Check test for possible deterioration of primers - Major defect - Code No. 09001.- If the total elapsed time between the original acceptance of any primer lot and the assembly of that lot into the primer base exceeds two years, or if the primers have been subjected to adverse conditions for any period of time since previous test, the primer lot shall be subjected to and must satisfactorily pass the check test for possible deterioration specified in Specification MIL-P-12951 immediately before the assembly of the primer lot into the primer bases. This test shall be performed by the contractor on primers selected by the Government inspector at the facility assembling the primer lot into the primer bases and shall be conducted at Government expense without cost to the contractor assembling the primers into the primer bases. Furthermore, this test shall not constitute a basis of rejection against either contractor, except where deterioration has occurred as a result of carelessness in handling, storage, etc., permitted while the primer lot was under the jurisdiction of either contractor.

4.3.3.3 Heat seal test of barrier bag (see dwg. 7548103) - Major defect - Code No. 06004.- Sampling and acceptance criteria for this test shall be in accordance with Specification MIL-P-116.

4.3.4 Inspection Equipment.- Equipment Tabulation Number ET-8822515 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.3.4.1 Government rights to documentation.- Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design,

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:39Z
Check the source to verify that this is the current version before use.

procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3.4.2 Supply and maintenance.- Supply and maintenance of the equipment listed in the ET shall be in accordance with Specification MIL-I-45607.

4.3.4.3 Government use of contractor's inspection equipment.- The contractor shall make available to the Government all inspection equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.4 Test methods and procedures

4.4.1 Functioning

4.4.1.1 Dud test.- Each primer base assembly shall be tested by dropping a steel ball weighing 1.94 plus or minus .02 ounces on the firing pin of the test fixture from a height not to exceed 13 inches. The assemblies shall comply with the requirements of 3.3.1 using test equipment specified in 4.3.4.

4.4.2 Heat seal test of barrier bag.- The test procedure for this test shall be as prescribed in Specification MIL-P-116 for the "Heat Seal Test".

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level C.- The primer and base assemblies shall be preserved, marked, and packaged in accordance with dwgs. 7548103 and 7548104.

5.2 Packing

5.2.1 Level C.- The primer and base assemblies shall be packed and marked in accordance with dwg. 9224760.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:39Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.
c. Submission of first article quantity.-Instructions as to the location of the facility at which the evaluation of the first article quantity shall be carried out, may be obtained from the contracting officer.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.7.

6.3.1 Equipment Lists.-Equipment part or identifying lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (SMUPA Form 1010) the numbers of drawings of existing equipment designs or codes as indicated in paragraph 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraphs 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:39Z
Check the source to verify that this is the current version before use.

6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3, 6.3.5 and 6.3.6 will be designated in the EL by the following codes. Other inspection equipment listed on the EL and not coded are defined in 6.3.4.

AR - Army design, mandatory for use.
CD - Contractor design and/or commercial equipment on contractor format. (Requires submission for Government approval as specified in 6.3.7).

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications, mandatory and non-mandatory. When an equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs (see 6.3.6) or commercial equipment selections (see 6.3.5) must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 Specification control drawing.-Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagrammatic or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings must be described in sufficient detail to permit identification and evaluation by the Government (see 6.3.5). Designs shall be submitted for approval as specified in 6.3.7.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:39Z
Check the source to verify that this is the current version before use.

6.3.5 Commercial equipment.-Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and Government. Contractors normally are not required to furnish drawings of commercial inspection equipment, but a list of such equipment must be approved by the Government. When commercial inspection equipment requires the use of special fixturing, the design of this fixturing shall be considered as part of the total design, and 6.3.6 shall apply. In the latter case, only those portions of the commercial equipment need be detailed in the designs as are necessary to completely fabricate, calibrate, and operate the total item of inspection equipment. Lists and fixturing designs shall be submitted for approval as specified in 6.3.7.

6.3.6 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finishes, notes, and operating and calibration instructic be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to Specification MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit one copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.7.

6.3.7 Submission of contractor designs.-Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commander, ARDEC, ATTN: SMCAR-QAT-M, Picatinny Arsenal N.J. 07806-5000. All other designs of inspection equipment shall

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:39Z
Check the source to verify that this is the current version before use.

be approved by the inspection element of the agency administering the contract; submission shall be as directed by the Contracting Officer. All submitted designs shall contain a reference to the applicable five digit Code Number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Submission of test data and ammunition data cards.- In addition to the normal distribution of records, for loaded items procured by the Department of the Army, one copy of all ballistic test data and ammunition data cards shall be forwarded to the following:

Commander:
ARDEC
ATTN: SMCAR-QAT-M
Picatinny Arsenal, NJ 07806-5000

6.5 Failure.- Defined as any one of the following primer malfunctions:

a. Misfire.- A failure wherein initiation of the primer mixture does not occur.
b. Squibbing.- A failure wherein the primer mixture burns with a fizz or hiss.
c. Hangfire.- A failure wherein an audible delay occurs between the instant of application of energy and the detonation report.

Custodian:
Army- AR

Preparing activity:
Army- AR

Project Number: 1375-A-1064

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:39Z
Check the source to verify that this is the current version before use.